

Taxonomy of Number

Number types • Factors • Primes • Indices • Standard Form • Surds

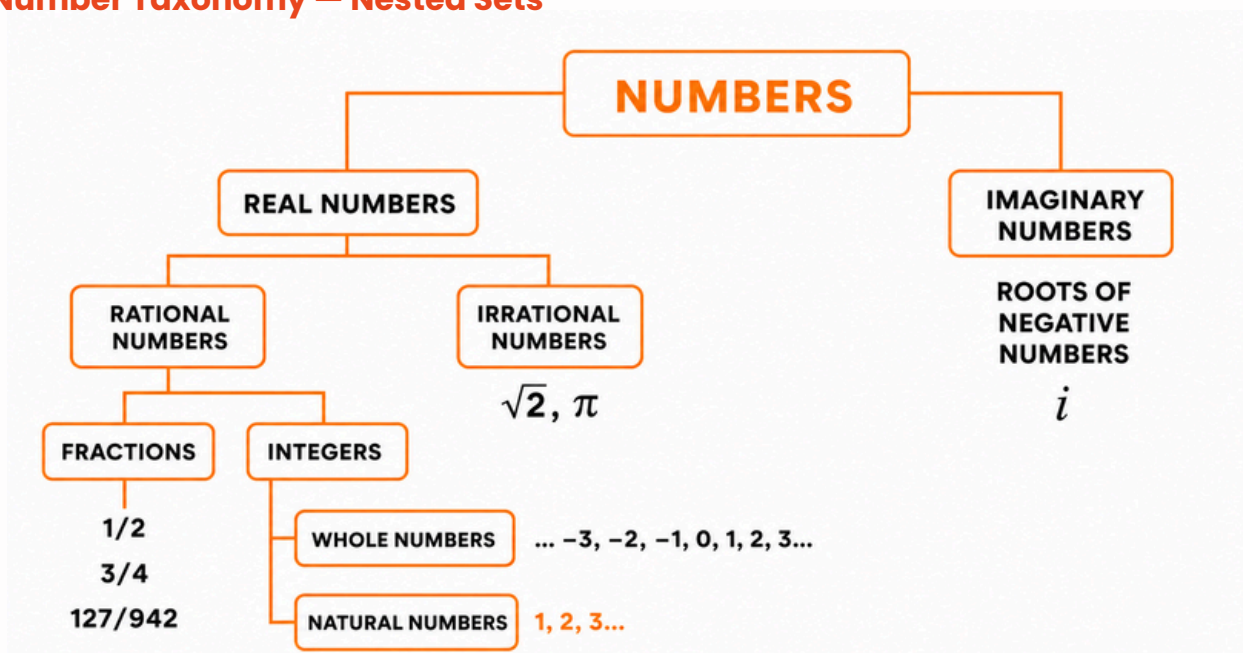


Use these notes well and you will be able to:

- Classify any number correctly: natural, integer, rational, irrational, real.
- Find HCF and LCM using prime factor decomposition.
- Apply all seven laws of indices (including negative and fractional).
- Write numbers in standard form and perform calculations with them.
- Simplify surds and rationalise the denominator.

If you follow the steps, you will succeed.

The Number Taxonomy – Nested Sets



Key Words

KEY WORD	
Whole numbers	Positive counting numbers: 1, 2, 3, 4, ...
Integers	All whole numbers: ... -3, -2, -1, 0, 1, 2, 3, ...
Rational numbers	Any number expressible as p/q (p, q integers, $q \neq 0$). Includes terminating and recurring decimals.
Irrational numbers	Cannot be expressed as a fraction. Non-terminating, non-recurring decimals. e.g. π , $\sqrt{2}$, e
Real numbers	All rational AND irrational numbers — everything on the number line.
Imaginary numbers	Complex numbers arising from the square root of negative 1
Prime numbers	Has exactly TWO factors: 1 and itself. e.g. 2, 3, 5, 7, 11...
Composite numbers	Has more than two factors — NOT prime. e.g. 4, 6, 8, 9...

Even & Odd numbers

Even numbers are numbers that can be divided by 2 without leaving a remainder.

Odd numbers are not divisible by two completely and always leave a remainder of 1

2	4
	6
	8
	10
	12
	14

Even numbers are the times table/multiples of 2

2	3	1
	5	1
	7	1
	9	1
	11	1
	13	1

Odd numbers end with 1,3,5,7 and 9.

Square & Cube numbers

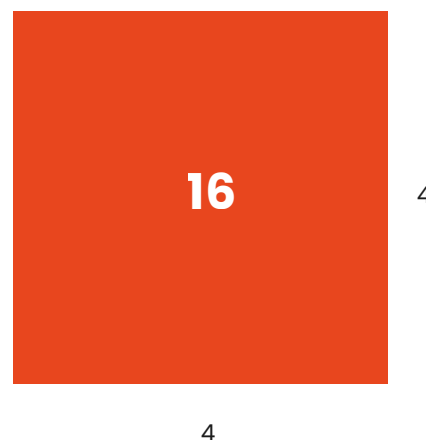
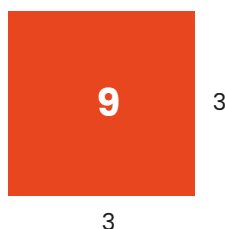
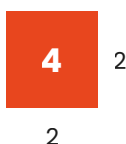
Square numbers are numbers you get when you multiply a whole number by itself.

For example:

- $1 \times 1 = 1$
- $2 \times 2 = 4$
- $3 \times 3 = 9$
- $4 \times 4 = 16$

So, 1, 4, 9, 16, 25, 36... are all square numbers.

They're called "square" numbers because you can arrange objects in a perfect square shape. For instance, 9 dots can form a 3 by 3 square.

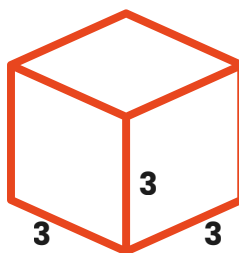


Note that the difference between the square numbers are all the odd numbers.

Cube numbers are numbers you get when you multiply a whole number by itself three times.

For example:

- $1 \times 1 \times 1 = 1$
- $2 \times 2 \times 2 = 8$
- $3 \times 3 \times 3 = 27$
- $4 \times 4 \times 4 = 64$



So, 1, 8, 27, 64, 125, 216... are cube numbers.

They're called "cube" numbers because you can arrange objects to form a perfect cube. For example, 27 small cubes can make a $3 \times 3 \times 3$ cube.

Prime Numbers

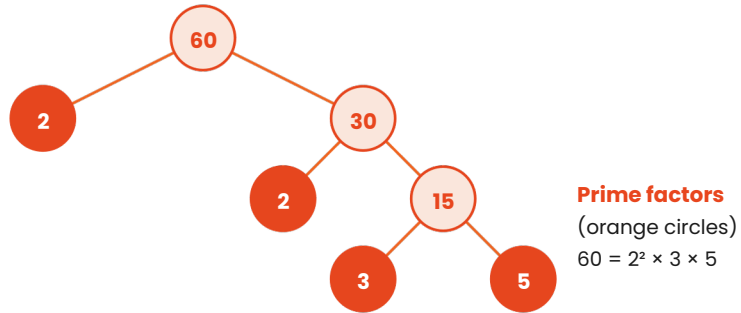
A prime number has EXACTLY two factors: 1 and itself. 1 is NOT prime (only one factor). 2 is the only EVEN prime. All other primes end in 1, 3, 7 or 9.

Primes up to 100: **2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97**

Prime Factor Decomposition — Two Methods

PRIME FACTORS, HCF & LCM

Factor Tree — $60 = 2^2 \times 3 \times 5$



HCF vs LCM

HCF (Highest Common Factor)

Find prime factors of both numbers.
Identify COMMON factors.
Multiply the LOWEST powers of common factors.
e.g. HCF(36, 48): $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$
Common: 2^2 and 3^1 HCF = $4 \times 3 = 12$

LCM (Lowest Common Multiple)

Find prime factors of both numbers.
Take ALL factors at their HIGHEST powers.
Multiply them together.
e.g. LCM(36, 48): take 2^4 and 3^2
LCM = $16 \times 9 = 144$

Find the HCF and LCM of 24 and 36.

$$24 = 2^3 \times 3 \quad 36 = 2^2 \times 3^2$$

HCF: lowest powers of common factors = $2^2 \times 3 = 12$

LCM: highest powers of ALL factors = $2^3 \times 3^2 = 72$

HCF = 12 | LCM = 72

INDICES (LAWS OF POWERS)

LAW	RULE	EXAMPLE
Multiplication	$a^x \times a^y = a^{x+y}$	$x^2 \times x = x^3$
Division	$a^x \div a^y = a^{x-y}$	$x \div x^2 = x^{-1}$
Power of power	$(a^x)^y = a^{xy}$	$(x^2)^3 = x^6$
Zero index	$a^0 = 1$	$7^0 = 1$
Negative index	$a^{-1} = 1/a$	$2^{-3} = 1/8$

Simplify: $(2x^3) \div (4x^2)$

Numerator: $2 \times x^{12} = 16x^{12}$

Divide: $16x^{12} \div 4x^2 = 4x^{10}$

Answer: $4x^{10}$

ADD YOUR NOTES HERE



Hey smart student!

Number taxonomy is the foundation of ALL of mathematics — get it right and everything else (algebra, geometry, statistics) builds on solid ground.

Proper classification of numbers in the right group is important

Keep calculating — see you in the next lesson!

From FREDi